Geometry & Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Circles: chords and arcs**

Points A and B lie on a circle of radius 5 centered at O, with central angle $\angle AOB = 90^\circ$. Find the length of chord AB, using $AB = 2r\sin\left(\frac{\theta}{2}\right)$.

$AB = 2(5)\sin(45^\circ) = 10(0.7071) \approx 7.07.$

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

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$AB = 2(5)\sin(45^\circ) = 10(0.7071) \approx 7.07.$

3 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $(1, 2)$ and $(4, 6)$.

$d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5.$

4 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of $(2, -3)$ and $(8, 5)$.

Midpoint $= \left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = (5, 1).$

5 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $(-2, 1)$ and $(3, -11)$.

$d = \sqrt{(3-(-2))^2 + (-11-1)^2} = \sqrt{25+144} = \sqrt{169} = 13.$

6 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of $(-4, 7)$ and $(6, -3)$.

Midpoint $= \left(\frac{-4+6}{2}, \frac{7-3}{2}\right) = (1, 2).$

7 **[Foundational]** • **Target: Gradient (slope)**

Find the gradient of the line through $(1, 2)$ and $(5, 10)$.

$m = \frac{10-2}{5-1} = \frac{8}{4} = 2.$

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Gradient (slope)

Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.

$m = \frac{-6-4}{2-(-3)} = \frac{-10}{5} = -2$.

9 [Intermediate] • Target: Gradient (slope)

Find the gradient of the line through $(0, 0)$ and $(6, 9)$.

$m = \frac{9-0}{6-0} = \frac{9}{6} = \frac{3}{2}$.

10 [Intermediate] • Target: Gradient (slope)

State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.

Both points have $y = 1$, so the line is horizontal (gradient $= 0$).

11 [Intermediate] • Target: Equations of lines

Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.

$y - 5 = 2(x - 3) \Rightarrow y = 2x - 1$.

12 [Intermediate] • Target: Equations of lines

Find the equation of the line through $(0, 4)$ and $(2, 10)$.

$m = \frac{10-4}{2-0} = 3$. Using the y-intercept 4: $y = 3x + 4$.

13 [Intermediate] • Target: Equations of lines

Find the x- and y-intercepts of the line $2x + 3y = 12$.

x-intercept ($y = 0$): $x = 6$. y-intercept ($x = 0$): $y = 4$.

14 [Intermediate] • Target: Equations of lines

Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.

Same gradient 4: $y - 9 = 4(x - 2) \Rightarrow y = 4x + 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 9.01 Statistics & Probability

State the gradient of a line perpendicular to a line with gradient 5.

$m = -\frac{1}{5}$.

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16 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 1.01 Algebra & Equations

Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.

Parallel — both have gradient 2.

17 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 2.01 Functions & Transformations

Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

The perpendicular gradient is 3. $y - 4 = 3(x - 3) \Rightarrow y = 3x - 5$.

18 [Challenging] • Target: The equation of a circle • Cross-Lesson: 3.01 Sequences & Series

Write the equation of a circle centered at the origin with radius 6.

$x^2 + y^2 = 36$.

19 [Challenging] • Target: The equation of a circle • Cross-Lesson: 4.01 Trigonometry

Write the equation of a circle centered at $(2, -3)$ with radius 5.

$(x - 2)^2 + (y + 3)^2 = 25$.

20 [Challenging] • Target: The equation of a circle • Cross-Lesson: 6.01 Limits & Derivatives

Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.

Center: $(1, -4)$. Radius: 7.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Distance and midpoint

Find the distance between the two points shown.

a. $\sqrt{5}$

b. 4

c. 3

d. 5 ✓

$\sqrt{(5 - 1)^2 + (5 - 2)^2} = \sqrt{16 + 9} = 5$.

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22 [Foundational] • Target: MCQ — Distance and midpoint

Find the midpoint of the segment joining the two points shown.

a. $(4, 3)$

b. $(2, 6)$

c. $(2, 3)$ ✓

d. $(8, 6)$

$\left(\frac{-2+6}{2}, \frac{3+3}{2}\right) = (2, 3)$.

23 [Foundational] • Target: MCQ — Equations of lines

The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.

a. $y = \frac{1}{2}x + 1$

b. $y = 2x + 1$ ✓

c. $y = x + 1$

d. $y = 2x - 1$

The line crosses the y-axis at 1 and rises 2 for every 1 across: $y = 2x + 1$.

24 [Foundational] • Target: MCQ — The equation of a circle

Write the equation of the circle shown, centered at the origin.

a. $x^2 + y^2 = 10$

b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$ ✓

$r = 5$; $r^2 = 25$.

25 [Foundational] • Target: MCQ — The equation of a circle

The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.

a. $(x - 1)^2 + (y - 2)^2 = 8$

b. $(x + 1)^2 + (y + 2)^2 = 16$

c. $(x - 1)^2 + (y - 2)^2 = 16$ ✓

d. $(x - 1)^2 + (y - 2)^2 = 4$

$(x - 1)^2 + (y - 2)^2 = 4^2$.

26 [Foundational] • Target: MCQ — Circles: chords and arcs

Find the arc length of the 90° sector shown, in terms of π.

a. 12π

b. 3π ✓

c. 1.5π

d. 6π

Arc $= \frac{90}{360} \times 2\pi(6) = 3\pi$.

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27 [Foundational] • Target: MCQ — Perimeter and area

A rectangle has length 15 and width 9, as shown. Find its perimeter and area.

a. $P = 135, A = 48$

b. $P = 48, A = 135$ ✓

c. $P = 24, A = 135$

d. $P = 48, A = 24$

$P = 2(15 + 9) = 48; A = 15 \times 9 = 135.$

28 [Foundational] • Target: MCQ — Perimeter and area

A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.

a. $A = 60$, hyp $= 13$

b. $A = 30$, hyp $= 17$

c. $A = 30$, **hyp** $= 13$ ✓

d. $A = 30$, hyp $= 12$

$A = \frac{1}{2}(5)(12) = 30$; hyp $= \sqrt{5^2 + 12^2} = 13.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Distance and midpoint

Find the distance between $(2, 3)$ and $(6, 6)$.

a. 5 ✓

b. 25

c. $\sqrt{7}$

d. 7

$\sqrt{(6-2)^2 + (6-3)^2} = \sqrt{16 + 9} = 5.$

30 [Intermediate] • Target: MCQ — Gradient (slope)

Find the gradient of the line through $(-2, 5)$ and $(3, -10)$.

a. $-\frac{1}{3}$

b. -15

c. -3 ✓

d. 3

$m = \frac{-10-5}{3-(-2)} = \frac{-15}{5} = -3.$

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31 [Intermediate] • Target: MCQ — Equations of lines

Find the x- and y-intercepts of the line $3x + 4y = 24$.

a. $x = 8, y = 6$ ✓

b. $x = 6, y = 8$

c. $x = 24, y = 24$

d. $x = 4, y = 3$

Set $y = 0$: $x = 8$. Set $x = 0$: $y = 6$.

32 [Intermediate] • Target: MCQ — Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = -\frac{1}{4}x + 3$, passing through $(4, 5)$.

a. $y = 4x - 5$

b. $y = 4x + 11$

c. $y = -\frac{1}{4}x + 6$

d. $y = 4x - 11$ ✓

Perpendicular gradient $= 4$: $5 = 4(4) + c \Rightarrow c = -11$.

33 [Intermediate] • Target: MCQ — The equation of a circle

Find the equation of the circle centered at the origin that passes through $(6, 8)$.

a. $x^2 + y^2 = 100$ ✓

b. $x^2 + y^2 = 10$

c. $x^2 + y^2 = 196$

d. $x^2 + y^2 = 48$

$r = \sqrt{6^2 + 8^2} = 10$; $r^2 = 100$.

34 [Intermediate] • Target: MCQ — Polygon angle sums

Find the measure of each interior angle of a regular hexagon.

a. 140°

b. 135°

c. 108°

d. 120° ✓

Sum $= 720^\circ$; each $= 720 \div 6 = 120^\circ$.

35 [Intermediate] • Target: MCQ — Coordinate geometry: identifying shapes

Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

a. No

b. Cannot be determined

c. **Yes** ✓

d. Only if it is also isosceles

AB is horizontal and BC is vertical, so they meet at a right angle at B: yes.

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36 [Intermediate] • Target: MCQ — The perpendicular bisector of a segment

A line segment has endpoints $(3, 6)$ and $(9, 6)$. Find the equation of its perpendicular bisector.

a. $x = 3$

b. $y = 6$

c. $y = 9$

d. $x = 6$ ✓

Midpoint $= (6, 6)$; the segment is horizontal, so the bisector is vertical: $x = 6$.

37 [Intermediate] • Target: MCQ — Circles: chords and arcs

Find the distance between $(3, 2)$ and $(-2, 14)$.

a. 15

b. 13 ✓

c. 17

d. 11

$d = \sqrt{(-2-3)^2 + (14-2)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Polygon angle sums • Cross-Lesson: 3.01 Sequences & Series

Find the midpoint of $(7, 6)$ and $(-9, 2)$.

a. $(0, 4)$

b. $(0, 5)$

c. $(-1, 5)$

d. $(-1, 4)$ ✓

Midpoint $= \left(\frac{7-9}{2}, \frac{6+2}{2}\right) = (-1, 4)$.

39 [Challenging] • Target: MCQ — Perimeter and area • Cross-Lesson: 4.01 Trigonometry

Find the gradient of the line through $(-3, 2)$ and $(-6, 3)$.

a. -3

b. $-\frac{1}{3}$ ✓

c. $\frac{1}{3}$

d. -1

$m = \frac{3-2}{-6-(-3)} = \frac{1}{-3} = -\frac{1}{3}$.

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40 [Challenging] • Target: MCQ — Coordinate geometry: identifying shapes • Cross-Lesson: 6.01 Limits & Derivatives

Find the equation of the line with gradient 4 passing through $(-4, 3)$, in the form $y = mx + c$.

a. $y = 4x + 21$

b. $y = 4x + 19$ ✓

c. $y = -4x + 19$

d. $y = 4x - 19$

$y - 3 = 4(x - (-4)) \Rightarrow y = 4x + 19$.

41 [Challenging] • Target: MCQ — The perpendicular bisector of a segment • Cross-Lesson: 7.01 Integration & Applications

State the gradient of a line perpendicular to a line with gradient $\frac{4}{3}$.

a. $-\frac{1}{2}$

b. $\frac{4}{3}$

c. $-\frac{3}{4}$ ✓

d. $\frac{3}{4}$

$m = -\frac{3}{4}$.

42 [Challenging] • Target: MCQ — Word problems: geometry on a coordinate grid • Cross-Lesson: 8.01 Exponentials & Logarithms

Write the equation of a circle centered at $(-6, -2)$ with radius 3.

a. $(x + 6)^2 + (y + 2)^2 = 16$

b. $(x - 6)^2 + (y + 2)^2 = 9$

c. $(x + 6)^2 + (y - 2)^2 = 9$

d. $(x + 6)^2 + (y + 2)^2 = 9$ ✓

$(x + 6)^2 + (y + 2)^2 = 9$.

43 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Statistics & Probability

Find the measure of each interior angle of a regular dodecagon.

a. 30°

b. 155°

c. 150° ✓

d. 145°

Sum $= (12 - 2)180^\circ = 1800^\circ$. Each angle $= \frac{1800^\circ}{12} = 150^\circ$.

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44 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Algebra & Equations

A right triangle has legs 12 and 16. Find its area and hypotenuse.

a. Area : 96. Hypotenuse : 21

b. Area : 96. Hypotenuse : 20 ✓

c. Area : 108. Hypotenuse : 20

d. Area : 192. Hypotenuse : 20

Area $= \frac{1}{2}(12)(16) = 96$. Hypotenuse $= \sqrt{144 + 256} = \sqrt{400} = 20$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions & Transformations

Find the equation of the perpendicular bisector of the segment joining $(4, 5)$ and $(-2, 13)$.

a. $y - 1 = \frac{3}{4}(x - 9)$

b. $y - 9 = \frac{3}{4}(x - 1)$ ✓

c. $y - 9 = -\frac{3}{4}(x - 1)$

d. $y - 9 = -\frac{4}{3}(x - 1)$

Midpoint $= (1, 9)$. Segment gradient $= -\frac{4}{3}$, so the perpendicular gradient is $\frac{3}{4}$. Equation: $y - 9 = \frac{3}{4}(x - 1)$.

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